

R Scenario Questions and Answers

Question 1 – IoT CSV

Scenario: IoT CSV

Task: Read daily CSV files, combine them, remove missing values and detect anomalies.

Answer:

Use `read.csv()` to read each CSV file, `list.files()` to list all CSV files, `rbind()` to combine them into one data frame, and `na.omit()` to remove missing values. Then analyze the cleaned data to detect anomalies.

Question 2 – Manufacturing Logs

Scenario: Manufacturing Logs

Task: Write cleaned production report to CSV and verify integrity.

Answer:

Use `write.csv()` to export the cleaned production report. Verify integrity by comparing the row count of the original cleaned data with the exported data after reading it back.

Question 3 – Network Logs

Scenario: Network Logs

Task: Import multiple log files and identify failed login patterns.

Answer:

Read all log files using `read.csv()` or `read.table()`, combine them using `rbind()`, and filter records where login status indicates failure to identify failed login patterns.

Question 4 – Weather Data

Scenario: Weather Data

Task: Read text and CSV formats and compare averages.

Answer:

Use `read.table()` to import text files, `read.csv()` for CSV files, and `summary()` or `mean()` to compare average values.

Question 5 – Smart City

Scenario: Smart City

Task: Export analyzed pollution report for management.

Answer:

Analyze the pollution data and export the final report using `write.csv()`.